

```
# This is formatted as code
```

for loops

```
for i in range(1,5):  
    print(i)
```

```
1  
2  
3  
4
```

```
for i in range(7,2,-1):  
    print(i)
```

```
7  
6  
5  
4  
3
```

```
for i in range(1,10):  
    if (i==5):  
        print(i)
```

```
5
```

```
for i in range(1,21):  
    if (i%2==0):  
        print("even",i)  
    elif True:  
        if (i%1==0):  
            print("number odd",i)  
    else:  
        print(i)
```

```
number odd 1  
even 2  
number odd 3  
even 4  
number odd 5  
even 6  
number odd 7  
even 8  
number odd 9  
even 10  
number odd 11  
even 12  
number odd 13  
even 14  
number odd 15  
even 16  
number odd 17
```

```
even 18
number odd 19
even 20
```

```
for i in range(57,19,-1):
    if (i%2==0 and i%5==0):
        print("divide",i)
    else:
        print("not divided",i)
```

```
not divided 57
not divided 56
not divided 55
not divided 54
not divided 53
not divided 52
not divided 51
divide 50
not divided 49
not divided 48
not divided 47
not divided 46
not divided 45
not divided 44
not divided 43
not divided 42
not divided 41
divide 40
not divided 39
not divided 38
not divided 37
not divided 36
not divided 35
not divided 34
not divided 33
not divided 32
not divided 31
divide 30
not divided 29
not divided 28
not divided 27
not divided 26
not divided 25
not divided 24
not divided 23
not divided 22
not divided 21
divide 20
```

```
t=0
for i in range (1,11):
    t = t+i
    print(t)
```

```
1
3
6
10
```

```
15
21
28
36
45
55
```

```
# when a loop from 3,37 print when number is divided by 3 and 4
for i in range(3,37):
    if (i%3==0 and i%4==0):
        print("divided by both",i)
    else:
        print("not divide ",i)
```

```
not divide 3
not divide 4
not divide 5
not divide 6
not divide 7
not divide 8
not divide 9
not divide 10
not divide 11
divided by both 12
not divide 13
not divide 14
not divide 15
not divide 16
not divide 17
not divide 18
not divide 19
not divide 20
not divide 21
not divide 22
not divide 23
divided by both 24
not divide 25
not divide 26
not divide 27
not divide 28
not divide 29
not divide 30
not divide 31
not divide 32
not divide 33
not divide 34
not divide 35
divided by both 36
```

```
# when a loop from 5,40 and print only odd number
for i in range (5,40):
    if (i%2==0):
        print("even",i)
    else:
        print("odd",i)
```

```
odd 5
even 6
odd 7
```

```
even 8
odd 9
even 10
odd 11
even 12
odd 13
even 14
odd 15
even 16
odd 17
even 18
odd 19
even 20
odd 21
even 22
odd 23
even 24
odd 25
even 26
odd 27
even 28
odd 29
even 30
odd 31
even 32
odd 33
even 34
odd 35
even 36
odd 37
even 38
odd 39
```

```
# when a loop 97,48 and print sum all the even numbers
s = 0
for i in range (97, 48, -1):
    if i %2 == 0:
        s = s+i
    print(s)
```

```
96
190
282
372
460
546
630
712
792
870
946
1020
1092
1162
1230
1296
1360
1422
1482
```

```
1540
1596
1650
1702
1752
```

```
data = "regex"
for char in data:
    print (char)
```

```
r
e
g
e
x
```

```
data = "divya "                # positive indexing
for char in range(0,5):
    print(data[2])
```

```
v
v
v
v
v
```

```
data = "divya "                # negative indexing
for char in data:
    print(data[-3])
```

```
y
y
y
y
y
y
y
```

```
# form 2,40 and 3 multification
for i in range (2,41):
    if i%3==0:
        print('divide',i)
    else:
        print("not divide",i)
```

```
not divide 2
divide 3
not divide 4
not divide 5
divide 6
not divide 7
not divide 8
divide 9
not divide 10
not divide 11
divide 12
not divide 13
```

```
not divide 14
divide 15
not divide 16
not divide 17
divide 18
not divide 19
not divide 20
divide 21
not divide 22
not divide 23
divide 24
not divide 25
not divide 26
divide 27
not divide 28
not divide 29
divide 30
not divide 31
not divide 32
divide 33
not divide 34
not divide 35
divide 36
not divide 37
not divide 38
divide 39
not divide 40
```

```
# 500,7 and multiply by 2, 3 ,8
for i in range (500,6,-1):
    if (i %2==0 and i % 3==0 and i%8==0):
        print ("divide",i)
    else:
        print("not divide",i)
```

```
not divide 39
not divide 38
not divide 37
not divide 36
not divide 35
not divide 34
not divide 33
not divide 32
not divide 31
not divide 30
not divide 29
not divide 28
not divide 27
not divide 26
not divide 25
divide 24
not divide 23
not divide 22
not divide 21
not divide 20
not divide 19
not divide 18
not divide 17
not divide 16
not divide 15
not divide 14
not divide 13
not divide 12
not divide 11
not divide 10
not divide 9
not divide 8
not divide 7
```